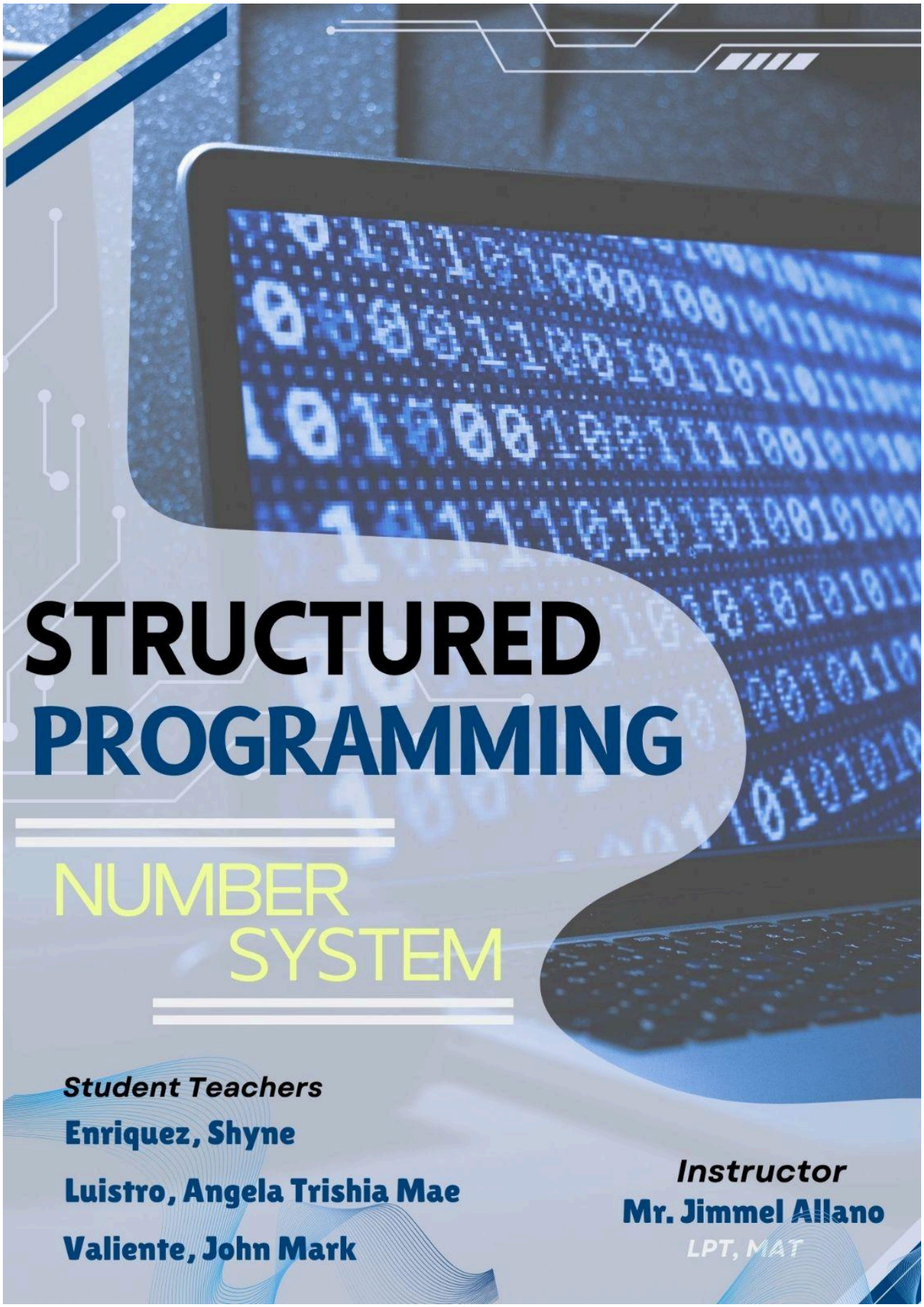
REPUBLIC OF THE PHILIPPINES
PHILIPPINE INSTITUTE OF COMPUTER PROGRAMMING (PICP)



STRUCTURED PROGRAMMING

NUMBER SYSTEM

Student Teachers

Enriquez, Shyne

Luistro, Angela Trishia Mae

Valiente, John Mark

Instructor

Mr. Jimmel Allano

LPT, MAT

REPUBLIC OF THE PHILIPPINES
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Structured Programming: Number System
Pre-test

Name: _____ Date:_____
Course, Year & Section : _____ Score:_____

Multiple Choice
Directions: Read each question carefully. Choose the letter of the **best** answer.

1. What is a number system?
A. A method of drawing numbers
B. A system used to represent numbers using digits or symbols
C. A program used to calculate numbers
D. A device used to display numbers

2. What is the base of the Binary Number System?
A. 2
B. 8
C. 10
D. 16

3. Which digits are used in the Binary Number System?
A. 0 and 1
B. 0 to 7
C. 0 to 9
D. 1 to 10

4. What is the base of the Hexadecimal Number System?
A. 2
B. 8
C. 10
D. 16

5. Why is the Decimal Number System called a base-10 system?
A. It has 10 different digits, from 0 to 9.
B. It can only represent 10 numbers.
C. It is used only for counting money.
D. It contains 10 binary digits.

6. Which statement best explains the Binary Number System?
A. It uses the digits 0–9.
B. It uses the digits 0–7.
C. It uses only 0 and 1.
D. It uses 0–9 and A–F.

7. Why are letters used in the hexadecimal system?
A. To make numbers look more complicated
B. To represent values from 10 to 15
C. To replace the numbers 0 to 9
D. To represent binary digits

8. Which statement correctly describes the four number systems?
A. Binary uses 2 digits, octal uses 8, decimal uses 10, and hexadecimal uses 16 symbols.
B. All four systems use the same digits.
C. Binary and decimal have the same base.
D. Octal has a higher base than hexadecimal.

9. A computer displays the number 10101_2. Which number system is being used?
A. Decimal
B. Binary
C. Octal
D. Hexadecimal

10. A student writes 58_8 as an octal number. Based on the digits allowed in octal, what should the student realize?
A. It is valid because octal uses 0–9.
B. It is invalid because 8 is not an octal digit.
C. It is valid because 8 is the base of octal.
D. It is invalid because octal uses only letters.

11. You are asked to write a number using hexadecimal symbols. Which choice follows the correct symbols?
A. 19G
B. 25H
C. 4AF
D. 78Z

12. A computer technician wants to represent a long binary value in a shorter form. Which number system would be useful for this purpose?
A. Decimal
B. Hexadecimal
C. Octal only

D. Counting numbers

13. Consider these numbers: 1011_2, 75_8, 84_{10}, and $3F_{16}$. What makes these numbers different from one another?
A. They use different number bases and sets of symbols.
B. They all use the same base but different symbols.
C. They are all binary numbers.
D. They represent completely different mathematical concepts.

14. A student says, “The number 19_8 is valid because 19 is a normal number in the decimal system.” What is the best analysis of the student's statement?
A. Correct, because all numbers are valid in every number system.
B. Correct, because octal uses all decimal digits.
C. Incorrect, because octal only allows digits 0 through 7.
D. Incorrect, because octal only uses 0 and 1.

15. Compare the numbers 101_2 and 101_{10}. Why can they have different values even though they contain the same digits?
A. Their bases are different.
B. Their symbols are different.
C. One is hexadecimal.
D. The digits have no value.

16. A computer file contains a long sequence of 0s and 1s. A programmer wants a shorter representation that is easier for humans to read while still being closely related to binary. Which system would be most appropriate?
A. Decimal
B. Hexadecimal
C. Roman numerals
D. Tally marks

17. A classmate says, “There is no reason to learn hexadecimal because decimal is easier for people to understand.” Which response best evaluates this claim?
A. The claim is completely correct because computers only use decimal.
B. The claim is reasonable for everyday counting, but hexadecimal is useful in computing because it provides a compact way of representing binary information.
C. The claim is correct because hexadecimal has no relationship with computers.
D. The claim is incorrect because people never use decimal numbers.

18. A student chooses Binary for everyday prices, Decimal for computer machine data, and Hexadecimal for representing long binary strings. Which part of the student's choices needs reconsideration?
A. Only the use of Decimal for computer machine data

B. Only the use of Binary for everyday prices
C. Only the use of Hexadecimal for long binary strings
D. All three choices are equally appropriate in every situation
Creating / Higher-Order Thinking

19. Imagine that you are designing a simple computer system that needs to represent only two conditions: "ON" and "OFF." Which representation would you choose and why?
A. Binary, because it can represent two states using 0 and 1.
B. Decimal, because it always uses exactly two digits.
C. Octal, because it only uses two symbols.
D. Hexadecimal, because it has the fewest symbols.

20. You are asked to explain to a younger student why computers use different number systems. Which explanation would be the most appropriate?
A. "Computers use different number systems randomly."
B. "Number systems provide different ways of representing values,
C. "Computers use hexadecimal because humans cannot understand decimal."
D. "All number systems have the same purpose and work exactly the same way."

"Your score reflects your learning, but your honesty reflects your character."

Prepared by:
Enriquez, Shynemark
Luistro, Angela Trishia Mae
Valiente, John Mark

ANSWER KEY

1. B
2. A
3. A
4. D
5. A
6. C
7. B
8. A
9. B
10. B
11. C
12. B
13. A
14. C
15. A
16. B
17. B
18. A
19. A
20. B

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MOTIVATIONAL ACTIVITY

"Can You Speak Computer?"

Directions: Read and complete the activity at your own pace.

"Think about the things you use every day"

- Your phone number
- The time displayed on your phone
- Your student number
- The price of your food
- The battery percentage of your phone
- Numbers displayed on a computer or calculator

1. Write three numbers that you commonly use in your daily life.

_____
_____
_____
_____

2. Where do you usually see these numbers?

_____
_____
_____
_____

3. Do you think a computer understands numbers exactly the same way humans do?

_____
_____
_____
_____